DTSC 2302 - Project 1 Part 2 - Group: Pavle D, Ikumi U, Sonia S, Michael F, and James H

Student Academic Performance Analysis and Predicting Final Grades Using OLS Regression

**Model Building Process**

To begin, a multiple linear regression model was developed to predict students' final grades (G3) using all available predictors except G1 and G2, as required. Since several variables were categorical, they were converted into dummy variables so they could be appropriately included in the regression model. The dataset was then split into an 80% training set and a 20% testing set to evaluate how well the model generalizes to new, unseen data and to reduce the risk of overfitting. Furthermore, the initial model included all predictors, so backward elimination was applied using p-values to remove the variables that were not statistically significant at the 0.05 level. This feature selection process improved the model's interpretability and reduced unnecessary noise by keeping only predictors that meaningfully contribute to explaining the variation in student performance.

**Model Description and Coefficient Interpretation**

The final model is a multiple linear regression that estimates G3 as a linear function of selected academic, behavioral, and demographic variables. The model assumes a linear relationship between the predictors and the final grade, allowing each variable to contribute independently to the predicted outcome while holding all other factors constant. After applying feature selection through backward elimination, only statistically significant predictors were retained in the final model, improving both interpretability and reliability. These remaining variables were found to have meaningful relationships with student performance and provide insight into the key factors influencing final grades.

For example, the coefficient for studytime (0.5311) indicates that, holding all other variables constant, a one-unit increase in study time is associated with an increase of approximately 0.53 points in the final grade. This suggests that increased time spent studying has a positive effect on academic performance.

Furthermore, the coefficient for failures (-1.3866) shows that each additional prior course failure is associated with a decrease of about 1.39 points in the final grade, holding other factors constant. This highlights the strong negative impact of prior academic struggles on future outcomes.

Lastly, the coefficient for higher_yes (1.9606) indicates that students who intend to pursue higher education score, on average, about 1.96 points higher than those who do not, holding all else constant. This suggests that academic motivation or future aspirations are positively associated with performance.

**Model Quality and Usefulness**

The final model demonstrates moderate explanatory and predictive performance, based on the standard evaluation metrics and overall model assessment. The in-sample R^2 is approximately 0.325, indicating that about 32.5% of the variation in final grades is explained by the predictors included in the model. When evaluated on the test dataset, the model achieves an R^2 of 0.2344, reflecting a decrease in performance on unseen data but still indicating reasonable generalization. Diagnostic checks were also considered, including examining residual behavior and potential multicollinearity, to validate the appropriateness of the linear regression model.

The test set RMSE of 2.2644 suggests that, on average, the model's predictions differ from actual final grades by approximately 2.26 points. While this level of error limits the model's precision for individual predictions, it is strong enough for identifying general trends and key factors associated with student performance.

Overall, the model is useful to an extent for understanding which variables are most strongly associated with academic outcomes, particularly study behavior, prior failures, and educational aspirations. However, the moderate R^2 indicates that a large portion of variability in student performance is driven by factors not included in the dataset. Furthermore, the reduced performance compared to Part 1 highlights the importance of prior academic performance variables, G1 and G2, as their exclusion likely contributes to the model's limited predictive power.

*Appendix:*

*Code*

```
import numpy as np
import pandas as pd
import statsmodels.api as sm
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_squared_error, r2_score

# load data
df = pd.read_csv("/Users/pavledimitrijevic/Downloads/student-performance_n_600.csv")
# drop unnecessary index column if it appears
if "Unnamed: 0" in df.columns:
   df = df.drop(columns=["Unnamed: 0"])
# define target and predictors
y = df["G3"]
X = df.drop(columns=["G3", "G1", "G2"])
# convert categorical variables to dummy variables
X = pd.get_dummies(X, drop_first=True)
# split into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(
   X, y, test_size=0.2, random_state=2302
)
# backward elimination for feature selection
while True:
   X_train_sm = sm.add_constant(X_train)
   model = sm.OLS(y_train, X_train_sm.astype(float)).fit()

   p_values = model.pvalues.drop("const")
   max_p = p_values.max()

   if max_p > 0.05:
       remove_var = p_values.idxmax()
       X_train = X_train.drop(columns=[remove_var])
       X_test = X_test.drop(columns=[remove_var])
   else:
       break
# fit final model
X_train_sm = sm.add_constant(X_train)
final_model = sm.OLS(y_train, X_train_sm.astype(float)).fit()
# predict on test set
X_test_sm = sm.add_constant(X_test, has_constant="add")
y_pred = final_model.predict(X_test_sm)
# evaluate out-of-sample performance
rmse = np.sqrt(mean_squared_error(y_test, y_pred))
r2 = r2_score(y_test, y_pred)
print("Final Model Test RMSE:", round(rmse, 4))
print("Final Model Test R^2:", round(r2, 4))
# model summary for appendix
print(final_model.summary())
```

*Model Summary*

```
Final Model Test RMSE: 2.2644
Final Model Test R^2: 0.2344
                            OLS Regression Results
==============================================================================
Dep. Variable:                     G3   R-squared:                       0.325
Model:                            OLS   Adj. R-squared:                  0.313
Method:                 Least Squares   F-statistic:                     28.29
Date:                Thu, 19 Mar 2026   Prob (F-statistic):           5.83e-36
Time:                        14:54:28   Log-Likelihood:                -1168.9
No. Observations:                 480   AIC:                             2356.
Df Residuals:                     471   BIC:                             2393.
Df Model:                           8
Covariance Type:            nonrobust
==================================================================================
                     coef    std err          t      P>|t|      [0.025      0.975]
----------------------------------------------------------------------------------
const             10.7590      0.696     15.448      0.000       9.390      12.128
Medu               0.2472      0.120      2.063      0.040       0.012       0.483
studytime          0.5311      0.165      3.216      0.001       0.207       0.856
failures          -1.3866      0.207     -6.697      0.000      -1.793      -0.980
Dalc              -0.4108      0.139     -2.962      0.003      -0.683      -0.138
health            -0.2012      0.090     -2.226      0.026      -0.379      -0.024
school_MS         -1.5470      0.284     -5.451      0.000      -2.105      -0.989
schoolsup_yes     -1.2183      0.403     -3.024      0.003      -2.010      -0.427
higher_yes         1.9606      0.439      4.469      0.000       1.098       2.823
...
==============================================================================
```